

Factorization of Matrices

Outline

LU factorization

Determinant from the PLU factorization

LU Factorization

Introduction: LU Factorization

- ▶ The LU factorization (or LU decomposition) expresses a matrix as the product of a **lower triangular** matrix and an **upper triangular** matrix.
- ▶ It is computed by Gaussian elimination: **U** is the final row echelon form, **L** stores the multipliers.
- ▶ When row swaps are needed, they are recorded in a permutation matrix **P** (PLU factorization).
- ▶ It is a fundamental tool to solve linear systems **$Ax = b$** efficiently.

PLU Factorization

Given a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, a **PLU factorization** of $\mathbf{A}$ is a decomposition of the form

$$\mathbf{PA} = \mathbf{LU},$$

where:

- ▶ $\mathbf{P} \in \mathbb{R}^{n \times n}$ is a **permutation matrix** (it accounts for possible row swaps),
- ▶ $\mathbf{L} \in \mathbb{R}^{n \times n}$ is a **lower triangular matrix** with ones on the diagonal,
- ▶ $\mathbf{U} \in \mathbb{R}^{n \times n}$ is an **upper triangular matrix**.

Key points:

- ▶ The factorization is computed by Gaussian elimination.
- ▶ If no row swaps are needed, then $P = I_n$ and the factorization reduces to the **LU factorization**:

$$A = LU.$$

- ▶ The PLU/LU factorization is very useful for solving linear systems $Ax = b$ efficiently.

Method to Compute PLU Factorization (1/2)

Method to compute the PLU (LU with pivoting) factorization.

Starting from $\mathbf{A}$, we perform Gaussian elimination.

Whenever a row swap is required, it is recorded in the permutation matrix $\mathbf{P}$.

At the end of the elimination we obtain the upper triangular matrix $\mathbf{U}$, while $\mathbf{L}$ stores the elimination multipliers.

Method to Compute PLU Factorization (2/2)

Specifically:

- ▶ Each elimination step $R_i \leftarrow R_i - \ell_{ij}R_j$ (with $i > j$) creates a zero in position (i, j) .
- ▶ The multiplier ℓ_{ij} is stored in position (i, j) of $\mathbf{L}$.
- ▶ The diagonal of $\mathbf{L}$ is filled with ones.
- ▶ If at step j rows i and k are swapped ($R_i \leftrightarrow R_k$), the multipliers already stored in rows i and k of $\mathbf{L}$ (in columns $1, \dots, j-1$) are swapped as well, and the same swap is applied to the rows of $\mathbf{P}$ (which starts as $\mathbf{I}_n$).

Example (1/5)

Let us compute the LU factorization of

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix}.$$

Step 1: Eliminate the entries below the first pivot $a_{11} = 2$.

We need to eliminate $a_{21} = 4$. The multiplier is:

$$\ell_{21} = \frac{a_{21}}{a_{11}} = \frac{4}{2} = 2.$$

Example (2/5)

We perform $R_2 \leftarrow R_2 - 2R_1$:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ -2 & 7 & 2 \end{pmatrix}.$$

We need to eliminate $a_{31} = -2$. The multiplier is:

$$\ell_{31} = \frac{a_{31}}{a_{11}} = \frac{-2}{2} = -1.$$

We perform $R_3 \leftarrow R_3 - (-1)R_1 = R_3 + R_1$:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{pmatrix}.$$

Example (3/5)

Step 2: Eliminate the entries below the second pivot -8 .

We need to eliminate the entry in position $(3, 2)$, which is currently 8 . The multiplier is:

$$\ell_{32} = \frac{8}{-8} = -1.$$

We perform $R_3 \leftarrow R_3 - (-1)R_2 = R_3 + R_2$:

$$\mathbf{U} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Example (4/5)

Step 3: Build the matrix $\mathbf{L}$.

The matrix $\mathbf{L}$ has ones on the diagonal and the multipliers below the diagonal:

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}.$$

Example (5/5)

Verification: We check that $\mathbf{LU} = \mathbf{A}$:

$$\begin{aligned}\mathbf{LU} &= \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 1 & 1 \\ 4 & 2-8 & 2-2 \\ -2 & -1+8 & -1+2+1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix} = \mathbf{A}. \checkmark\end{aligned}$$

When Is a Row Swap Needed?

Gaussian elimination may stop because a pivot is zero.

We now characterize the matrices for which the LU factorization exists **without row swaps**.

Leading Principal Minors

- ▶ Given $\mathbf{A} \in \mathbb{R}^{n \times n}$ and $k \in \{1, \dots, n\}$, we denote by $\mathbf{A}_{[k]}$ the **leading principal submatrix** of order k , i.e., the $k \times k$ matrix formed by the first k rows and the first k columns of $\mathbf{A}$.
- ▶ Its determinant $\det(\mathbf{A}_{[k]})$ is called the **leading principal minor** of order k .
- ▶ Note that $\mathbf{A}_{[n]} = \mathbf{A}$.

Observation 1: Existence of the LU factorization without row swaps

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a nonsingular matrix. Then $\mathbf{A}$ admits a factorization $\mathbf{A} = \mathbf{L}\mathbf{U}$, with $\mathbf{L}$ lower triangular with ones on the diagonal and $\mathbf{U}$ upper triangular, if and only if all its leading principal minors are nonzero:

$$\det(\mathbf{A}_{[k]}) \neq 0, \quad \forall k \in \{1, \dots, n\}.$$

In this case, Gaussian elimination never requires a row swap and the pivots are

$$u_{11} = \det(\mathbf{A}_{[1]}), \quad u_{kk} = \frac{\det(\mathbf{A}_{[k]})}{\det(\mathbf{A}_{[k-1]})}, \quad k \in \{2, \dots, n\}.$$

Proof (1/2)

($\Leftarrow$) We perform Gaussian elimination without row swaps.

- ▶ A row addition $R_i \leftarrow R_i - \ell_{ij}R_j$ with $j < i$ does not change any leading principal minor: if $k \geq i$, it acts on $\mathbf{A}_{[k]}$ as a row addition; if $k < i$, it does not modify $\mathbf{A}_{[k]}$.
- ▶ Suppose that the first $k - 1$ steps have been performed. Then the leading principal submatrix of order k of the current matrix is upper triangular with diagonal entries $u_{11}, \dots, u_{kk}$, hence

$$\det(\mathbf{A}_{[k]}) = u_{11} u_{22} \cdots u_{kk}.$$

- ▶ Since $\det(\mathbf{A}_{[k]}) \neq 0$, the pivot $u_{kk} = \det(\mathbf{A}_{[k]}) / \det(\mathbf{A}_{[k-1]})$ is nonzero and step k can be performed without a row swap.

Proof (2/2)

($\Rightarrow$) Let $\mathbf{A} = \mathbf{L}\mathbf{U}$.

- ▶ Since $\mathbf{L}$ is lower triangular and $\mathbf{U}$ is upper triangular, the first k rows and columns of the product only involve the first k rows and columns of the factors:

$$\mathbf{A}_{[k]} = \mathbf{L}_{[k]} \mathbf{U}_{[k]}.$$

- ▶ Hence

$$\det(\mathbf{A}_{[k]}) = 1 \cdot u_{11} \cdots u_{kk}.$$

- ▶ Since $\det(\mathbf{A}) = u_{11} \cdots u_{nn} \neq 0$, all u_{ii} are nonzero, and so are all the leading principal minors.



Leading Principal Minors and Pivots

- For the matrix of the previous example (LU factorization), we have

$$\det(\mathbf{A}_{[1]}) = 2, \quad \det(\mathbf{A}_{[2]}) = 2 \cdot (-6) - 1 \cdot 4 = -16, \quad \det(\mathbf{A}_{[3]}) = \det(\mathbf{A}) = -16,$$

which give the pivots

$$u_{11} = 2, \quad u_{22} = -16/2 = -8, \quad u_{33} = -16/(-16) = 1.$$

- If some leading principal minor is zero, a row swap is needed and we must compute a PLU factorization (with $\mathbf{P} \neq \mathbf{I}$). This happens in the next examples.

Example (1/3)

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Since $a_{11} = 0$, we cannot use it as a pivot. We need to swap rows 1 and 2.

Let

$$\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

be the permutation matrix that swaps rows 1 and 2.

Example (2/3)

Then:

$$\mathbf{PA} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Now $\mathbf{PA}$ already has an upper triangular form (no elimination needed below the diagonal), so:

$$\mathbf{L} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}, \quad \mathbf{U} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Example (3/3)

Therefore, the PLU factorization is:

$$PA = LU,$$

where

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad L = I, \quad U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Example (1/5)

Let us compute the PLU factorization of

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 2 & 3 \\ -2 & 3 & 1 \end{pmatrix}.$$

Note that $\det(\mathbf{A}_{[2]}) = 2 \cdot 2 - 1 \cdot 4 = 0$: a row swap will be needed.

Step 1: Eliminate the entries below the first pivot $a_{11} = 2$. The multipliers are

$$\ell_{21} = \frac{4}{2} = 2, \quad \ell_{31} = \frac{-2}{2} = -1.$$

Example (2/5)

Step 1 (continued):

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 0 & 1 \\ -2 & 3 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 4 & 2 \end{pmatrix} \quad (R_3 \leftarrow R_3 + R_1)$$

Example (3/5)

Step 2: The entry in position $(2, 2)$ is 0 and cannot be used as a pivot. We swap rows 2 and 3:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & 1 \end{pmatrix} \quad (R_2 \leftrightarrow R_3)$$

The multipliers already computed for rows 2 and 3 are swapped as well: now $\ell_{21} = -1$ and $\ell_{31} = 2$. The entry below the new pivot 4 is already zero, so $\ell_{32} = 0$, and the elimination is complete:

$$\mathbf{U} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Example (4/5)

Step 3: Build $\mathbf{P}$ and $\mathbf{L}$. The matrix $\mathbf{P}$ is obtained from $\mathbf{I}_3$ by swapping rows 2 and 3, and $\mathbf{L}$ contains the (swapped) multipliers:

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}.$$

Example (5/5)

Verification: We check that $\mathbf{PA} = \mathbf{LU}$:

$$\mathbf{PA} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 4 & 2 & 3 \\ -2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ -2 & 3 & 1 \\ 4 & 2 & 3 \end{pmatrix},$$

$$\mathbf{LU} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ -2 & -1+4 & -1+2 \\ 4 & 2 & 2+1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ -2 & 3 & 1 \\ 4 & 2 & 3 \end{pmatrix} = \mathbf{PA}.$$

Why the Multipliers Must Be Swapped

- ▶ Without swapping the multipliers in Step 2 we would get the wrong matrix $\tilde{\mathbf{L}}$ with

$$\tilde{\ell}_{21} = 2, \quad \tilde{\ell}_{31} = -1,$$

and $\tilde{\mathbf{L}}\mathbf{U} \neq \mathbf{PA}$ (its second row would be (4, 6, 4)).

- ▶ Every row swap must be applied consistently to $\mathbf{P}$, to the current matrix, and to the multipliers already stored in $\mathbf{L}$.

Summary (1/2)

PLU factorization:

- ▶ $\mathbf{PA} = \mathbf{LU}$: $\mathbf{P}$ permutation, $\mathbf{L}$ lower triangular with unit diagonal, $\mathbf{U}$ upper triangular
- ▶ Computed by Gaussian elimination: $\mathbf{U}$ is the final matrix, ℓ_{ij} stored in $\mathbf{L}$
- ▶ A row swap is applied to $\mathbf{P}$ and to the multipliers already stored in $\mathbf{L}$

LU factorization ($\mathbf{P} = \mathbf{I}$):

- ▶ For $\mathbf{A}$ nonsingular, it exists if and only if $\det(\mathbf{A}_{[k]}) \neq 0$ for all k
- ▶ Pivots: $u_{kk} = \det(\mathbf{A}_{[k]}) / \det(\mathbf{A}_{[k-1]})$

Summary (2/2)

Examples:

- ▶ 3×3 LU factorization without row swaps
- ▶ 2×2 PLU factorization: zero pivot a_{11} , swap at the first step
- ▶ 3×3 PLU factorization: zero pivot in position $(2, 2)$, swap during the elimination

Determinant from the PLU Factorization

Introduction: Determinant from the PLU Factorization

- ▶ Once a PLU factorization is available, the determinant of $\mathbf{A}$ can be read almost for free.
- ▶ We only need the diagonal of $\mathbf{U}$ and the number of row swaps.
- ▶ This also gives a simple test of invertibility.

Observation 2: Determinant from the PLU factorization

Let $\mathbf{PA} = \mathbf{LU}$ be a PLU factorization of $\mathbf{A} \in \mathbb{R}^{n \times n}$, where $\mathbf{P}$ is obtained from $\mathbf{I}_n$ with s row swaps. Then

$$\det(\mathbf{A}) = (-1)^s \prod_{i=1}^n u_{ii} = \pm \prod_{i=1}^n u_{ii}.$$

In particular, $\mathbf{A}$ is invertible if and only if all the diagonal entries of $\mathbf{U}$ are nonzero.

Proof

- ▶ Each row swap changes the sign of the determinant, so $\det(\mathbf{P}) = (-1)^s \det(\mathbf{I}_n) = (-1)^s$.
- ▶ The matrices $\mathbf{L}$ and $\mathbf{U}$ are triangular, so $\det(\mathbf{L}) = 1$ and $\det(\mathbf{U}) = \prod_{i=1}^n u_{ii}$.
- ▶ From $\det(\mathbf{PA}) = \det(\mathbf{LU})$ and the product rule of determinants:

$$\det(\mathbf{P}) \det(\mathbf{A}) = \det(\mathbf{L}) \det(\mathbf{U})$$

$$(-1)^s \det(\mathbf{A}) = \prod_{i=1}^n u_{ii},$$

and we multiply both sides by $(-1)^s$.



Example

- ▶ LU example: no row swaps ($s = 0$), so $\det(\mathbf{A}) = 2 \cdot (-8) \cdot 1 = -16$.
- ▶ 2×2 PLU example: one row swap ($s = 1$), so $\det(\mathbf{A}) = -(1 \cdot 1) = -1$.
Indeed, $\det(\mathbf{A}) = 0 \cdot 1 - 1 \cdot 1 = -1$.
- ▶ 3×3 PLU example: one row swap ($s = 1$), so $\det(\mathbf{A}) = -(2 \cdot 4 \cdot 1) = -8$.
Indeed, by Laplace expansion along the first row,

$$\det(\mathbf{A}) = 2 \cdot (2 - 9) - 1 \cdot (4 + 6) + 1 \cdot (12 + 4) = -14 - 10 + 16 = -8.$$

Summary

Determinant from PLU:

- ▶ $\det(\mathbf{A}) = (-1)^s \prod_{i=1}^n u_{ii}$, with s the number of row swaps
- ▶ $\det(\mathbf{P}) = (-1)^s$, $\det(\mathbf{L}) = 1$, $\det(\mathbf{U}) = \prod_i u_{ii}$
- ▶ $\mathbf{A}$ is invertible if and only if all $u_{ii} \neq 0$

Key Takeaways

Summary

1. PLU Factorization:

- ▶ Decomposition of $\mathbf{A}$ into $\mathbf{PA} = \mathbf{LU}$
- ▶ $\mathbf{P}$: permutation matrix (row swaps); $\mathbf{L}$: lower triangular with ones on diagonal; $\mathbf{U}$: upper triangular

2. Method:

- ▶ Computed via Gaussian elimination, multipliers ℓ_{ij} stored in $\mathbf{L}$
- ▶ If no pivoting needed: $\mathbf{P} = \mathbf{I}$ and we get LU factorization (iff all leading principal minors are nonzero, for $\mathbf{A}$ nonsingular)

3. Applications:

- ▶ Efficient solution of linear systems $\mathbf{Ax} = \mathbf{b}$
- ▶ Determinant: $\det(\mathbf{A}) = (-1)^s \prod_i u_{ii}$; matrix inversion